

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I S.PRAVEEN (192524239) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ARTICLE REVIEW

Course: Artificial Intelligence (CSA17) | CO4 | Name: S.PRAVEEN | Reg No: 192524239

Task 1: Article Summary

(a) Citation, Domain, and ML Problem

Das, A., Choudhury, D., & Sen, A. (2023). A collaborative empirical analysis on machine learning based disease prediction in health care system. International Journal of Information Technology, 16(1), 261–270. DOI: <https://doi.org/10.1007/s41870-023-01556-5>

Domain / Application Area: Healthcare informatics – clinical decision support for preliminary, symptom-based disease diagnosis.

Specific ML Problem Addressed: A multi-class classification problem in which the most likely disease (out of roughly forty possible disease categories) is predicted from a patient's set of reported symptoms, using supervised learning.

(b) Key Objective and Research Gap (in own words)

This study addresses the growing need for AI-assisted preliminary diagnosis support in healthcare by comparing four widely used supervised classifiers – K-Nearest Neighbour, Naive Bayes, Decision Tree, and Random Forest – on their ability to predict a disease from a set of reported symptoms. The authors observe that although many prior works apply individual machine learning models to disease prediction, most such studies evaluate only one or two algorithms in isolation and rarely benchmark them under identical conditions. The research gap identified is therefore twofold: first, the lack of a unified, head-to-head empirical comparison of an instance-based, a probabilistic, and two tree-based classifiers on the same symptom dataset; and second, the absence of an accessible, end-user-facing tool that translates model predictions into a usable diagnostic aid. To fill this gap, the authors trained all four models on the same publicly available symptom–disease dataset, benchmarked them using consistent accuracy metrics, and packaged the best-performing model into a web-based application that lets a user select symptoms and receive an instant predicted diagnosis.

Task 2: Methodology Analysis

(a) ML Technique(s) and Dataset

The paper implements and compares four supervised classifiers: **K-Nearest Neighbour** (an instance-based, lazy learner that classifies a new case by majority vote among its nearest labelled neighbours), **Naive Bayes** (a probabilistic classifier built on Bayes' theorem under a conditional-independence assumption between features), **Decision Tree** (a recursive, attribute-splitting tree classifier), and **Random Forest** (a bagging ensemble of many decision trees).

The dataset used is the publicly available Kaggle "Disease Prediction Using Machine Learning" dataset originally compiled by Patil (2022), which is commonly documented as containing a few thousand patient records, each represented as a binary vector across roughly 130 possible symptoms and labelled with one of about forty distinct disease classes (for example, common cold, diabetes, malaria, and hepatitis). Because the features are pre-encoded as binary symptom-presence flags, minimal additional pre-processing was needed beyond splitting the data into training and testing partitions and

encoding the categorical disease labels; all four algorithms were then trained on the same split so that performance differences could be attributed to the choice of model rather than to differences in data handling.

(b) Mapping to CO4, Strength, and Limitation

Mapping to CO4: the Decision Tree component maps directly to the Decision Tree Algorithms topic of CO4, the Naive Bayes component maps to the Statistical Learning Methods topic, and the overall approach of learning a general symptom-to-disease mapping from a finite set of labelled training examples is a direct application of Inductive Learning.

Methodological strength: benchmarking four conceptually different classifiers – a lazy instance-based method, a probabilistic method, a single tree, and a tree ensemble – on identical data and identical evaluation metrics gives a fair, reproducible comparison and allows the ensemble's accuracy gain over the single Decision Tree to be quantified directly rather than inferred from separate studies.

Methodological limitation: the paper does not appear to report class-wise performance (precision/recall per disease), cross-validation results, or an assessment of how the models perform on noisy or incomplete real-world symptom reports – all of which matter more in a genuine clinical deployment than the single aggregate accuracy figure highlighted for each model.

Task 3: Results & Critical Evaluation

(a) Key Results

The authors report the following classification accuracies, benchmarked on the same symptom–disease dataset:

Model	Type	Reported Accuracy
Naive Bayes	Statistical / Probabilistic	87.7%
K-Nearest Neighbour	Instance-based	93.4%
Decision Tree	Tree-based	95.8%
Random Forest	Ensemble (bagged trees)	99.5%

Random Forest achieved the highest accuracy, followed by Decision Tree, K-Nearest Neighbour, and Naive Bayes, indicating that the tree-ensemble method generalised best on this dataset while the single Naive Bayes model, constrained by its independence assumption, performed weakest.

(b) Critical Evaluation

The reported accuracies, particularly Random Forest's 99.5% and Decision Tree's 95.8%, are high enough to raise questions about how realistic they are for a genuine diagnostic setting. Symptom–disease benchmark datasets of this kind are often curated so that each disease maps to a fairly distinct, non-overlapping symptom signature, which makes classification easier than it would be with real patient data, where symptoms overlap heavily across diseases and are frequently under-reported or inconsistently recorded. Potential biases include: (i) *selection bias*, since the dataset

was assembled for a public benchmark rather than collected from an actual clinical population; (ii) *evaluation bias*, since it is not clear whether k-fold cross-validation was used or whether the test set was a single held-out split, which affects how much the reported numbers can be trusted; and (iii) *class-balance bias*, since it is unclear whether every disease category is represented by a similar number of samples. Additional experiments that would strengthen the findings include k-fold cross-validation with reported variance, a confusion matrix broken down by disease, testing on an independent, hospital-sourced dataset, and a comparison against a simple majority-class baseline to show how much of the reported accuracy is attributable to genuine learning rather than to the structure of the dataset itself.

(c) Real-World Applicability

A symptom-to-disease classifier of this kind is well suited to low-stakes, preliminary triage applications: a self-assessment chatbot that suggests when a patient should see a doctor, a first-pass screening tool in a telemedicine app, or a clinical decision-support widget that helps a nurse prioritise which patients to see first. It is not suited, on its own, to final diagnosis, since an incorrect prediction in a medical context carries real consequences. Scaling this to an industry setting faces several challenges: real clinical data is far messier and sparser than the clean benchmark dataset used here; gathering enough labelled symptom–diagnosis pairs at scale raises patient privacy and consent issues (HIPAA/GDPR-type constraints); the compute and validation cost of retraining models on much larger, continuously updated datasets is non-trivial for smaller healthcare providers; and there are genuine ethical risks around over-reliance on an opaque model's output, especially for patient groups whose symptom presentation differs from the training population.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights Linked to CO4

- 1 **Ensemble methods outperform their base learner in a measurable, reproducible way.** Seeing Random Forest gain roughly 3.7 percentage points of accuracy over a single Decision Tree on the same data made the idea of bagging – combining many variance-prone trees to reduce overall variance – concrete. This connects directly to the CO4 topic of Decision Tree algorithms and their ensemble extensions.
- 2 **A model's assumptions shape where it fails, not just how well it scores on average.** Naive Bayes had the lowest accuracy of the four models, consistent with its conditional-independence assumption being violated when symptoms co-occur, as they typically do in real diseases. This reinforced the Statistical Learning Methods portion of CO4, specifically why Naive Bayes is fast and interpretable but weaker when features are correlated.
- 3 **High accuracy on a benchmark dataset is not the same as being ready for deployment.** Comparing the models' reported numbers against how the dataset was constructed made clear that Inductive Learning generalises only as well as the training distribution represents the real-world distribution it will be applied to – a reminder that the CO4 goal of "building models for classification and decision-making" must include evaluating whether the training data is representative, not just optimising accuracy.

(b) Proposed Extension

A worthwhile extension would be to replace the Decision Tree and Random Forest baselines with a gradient-boosted tree model (e.g., XGBoost or LightGBM) combined with a SHAP-based explainability layer, and to evaluate all models not only on the original Kaggle benchmark but also on a second, independently sourced and deliberately noisier symptom dataset (with missing values and symptom overlap deliberately introduced) to test robustness. This is feasible with existing open-source libraries and modest computational resources, since gradient boosting and SHAP are both mature, well-documented tools that do not require specialised hardware. The expected impact is twofold: gradient boosting typically edges out Random Forest on tabular classification tasks of this kind, and the SHAP layer would let a clinician see which reported symptoms drove a given prediction – directly addressing the interpretability gap that is essential for any real clinical adoption of an inductive learning-based diagnostic aid.